

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Interdisciplinary Project Work Report on

“SkillMatch-

A Skill Matching Platform”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the
Mini Project under the Inter Disciplinary Project,
Department of Basic Sciences.

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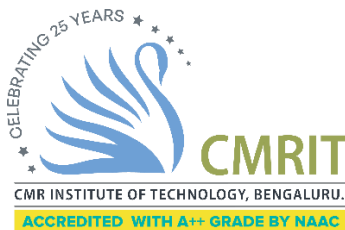
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CERTIFICATE

Certified that the project work entitled “SkillMatch – A Skill matching platform” is a bonafide work carried out by:

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Bonafide students of **CMR Institute of Technology**. The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Inter Disciplinary Project, Department of Basic Sciences**. It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract

SkillMatch is an AI-powered job recommendation and skill verification platform designed to improve the connection between students and employment opportunities. Traditional resume-based recruitment systems often fail to accurately evaluate practical skills and relevant competencies. The proposed system addresses this limitation by combining resume parsing, skill extraction, semantic similarity search, and skill-based assessments to provide more accurate job recommendations. The platform uses Natural Language Processing techniques to extract skills from resumes and compares them with job requirements using embedding-based similarity matching and FAISS search. A web-based interface was developed to allow students to upload resumes, take skill tests, track progress, and view personalized recommendations. Employers can evaluate candidates based on verified skills rather than keyword matching alone. Experimental testing demonstrated improved recommendation relevance, faster matching performance, and efficient resume analysis. User feedback indicated that the system effectively identified suitable opportunities and highlighted skill gaps for improvement. The project successfully demonstrates the potential of AI-driven recruitment systems in enhancing skill-based hiring and career guidance.

Keywords:

Artificial Intelligence, Resume Parsing, Skill Matching, Job Recommendation System, Natural Language Processing, FAISS Search

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Chapter 1

INTRODUCTION

It goes without saying that today's job search space is full of noise. Prospective candidates spend many hours looking through postings unrelated to their field of expertise, while recruiters go through hundreds of resumes which don't even come close to being a match for the position. However, the discrepancy between what candidates know and what employers are looking for constantly increases, and no proper solution yet exists.

And that's exactly where SkillMatch comes in. This is an AI-driven platform that scans the resume submitted by a job seeker, extracts their competencies in semantic space, and identifies job positions that might be a perfect fit. From the employer's perspective, the platform works in the opposite direction – analyzes the description of the job opening and finds the candidate with suitable abilities.

To put it briefly, this concept is based on matching semantics rather than titles and keywords. Large Language Model (LLM) embeddings are used to convert both the skill and requirement into vectors within semantic space, and then these vectors are matched by similarity.

1.1 Background

There have been huge advances in job discovery due to the availability of job boards such as LinkedIn, Naukri, and Indeed; however, there have been no improvements in terms of quality matching. This is because keyword search does not understand that building REST APIs in Flask is the same thing as Python back-end development. Candidates feel lost, and the process takes forever.

However, thanks to the advances in the NLP area, particularly in embedding based on transformer-based sentences, text can be semantically represented within seconds using commodity machines.

1.2 Modern AI Matching

The two foundations upon which SkillMatch was built are semantic search and vector databases. Free text skill descriptions are transformed to vectors through sentence transformers. FAISS (or a similar approximate nearest neighbor library) creates indexes for those vectors, providing sub-millisecond lookups. The end product is a matching engine that understands synonyms and related concepts without needing to be tagged manually.

Chapter 2

LITERATURE SURVEY

SkillMatch — The Freshman Fast-Track Job Portal

Paper (1): Skills-Based Hiring: The Long Road from Pronouncements to Practice

Publisher, Author, Year: Burning Glass Institute, Sigelman, M., Fuller, J., Martin, A., 2024

This report by the Burning Glass Institute analyzes a large number of real job postings to examine whether companies have truly adopted skills-based hiring or are still relying on degree filters. The authors found that degrees are frequently used as a shortcut screening tool, even for roles where a degree does not directly predict job performance. Notably, around 62% of candidates without a formal degree are filtered out of jobs they are actually capable of performing. The report concludes that while many organizations claim to have moved toward skills-based hiring, actual implementation in job postings and hiring pipelines remains inconsistent. It highlights a significant gap between employer intent and organizational practice.

Relevance: SkillMatch directly addresses this gap by replacing degree-based filtering with task and badge-based evaluation. This paper validates our core premise that resume and degree-based hiring unfairly excludes capable candidates, particularly first-year students who have not yet completed a degree but may already possess practical skills.

Paper (2): The State of Skills-Based Hiring 2023

Publisher, Author, Year: TestGorilla, TestGorilla Research Team, 2023

This industry report surveyed thousands of employers across multiple countries about their experience with skills-based hiring versus traditional resume screening. A major finding was that over 50% of employers found resume accuracy to be a serious concern, and 92% of respondents believed that skills assessments are more effective than resumes for identifying qualified candidates. The report also documented a growing trend of companies replacing or supplementing resumes with practical tests. It further found that skills-based hiring reduces mis-hires, decreases time-to-hire, and improves long-term employee retention. The data provides strong empirical support for moving away from resume-first evaluation frameworks.

Relevance: This report forms the empirical backbone of SkillMatch's design philosophy. Since 92% of employers prefer skills assessments over resumes, our platform's challenge-based application system — where students complete real tasks instead of submitting resumes — is directly aligned with what employers actually want. It also supports our badge-based credentialing as a more trustworthy signal of candidate capability.

Paper (3): A Smart Secured Framework for Detecting and Averting Online Recruitment Fraud

Publisher, Author, Year: PeerJ Computer Science, Ullah, Z. & Jamjoom, M., 2023

This research paper presents a machine learning-based framework designed to detect fraudulent job postings and fake applicant profiles on online recruitment platforms. The authors used ensemble learning methods, combining multiple classifiers to improve detection accuracy. Their study revealed that online recruitment fraud is widespread, affecting both job seekers who fall for fake postings and employers who receive fraudulent applications. The paper discusses the types of deceptive content found in recruitment, including misleading job descriptions and inflated applicant credentials. The proposed system was evaluated on real-world datasets and showed strong performance in flagging suspicious listings and profiles with high precision.

Relevance: This paper informed SkillMatch's approach to resume fraud detection. While the methods described are more advanced than what we currently implement, it helped us understand the scale of recruitment fraud and the types of patterns that are detectable through automated systems. SkillMatch's basic duplicate-detection and similarity-checking logic is inspired by the principles of text-based fraud detection outlined in this work.

Paper (4): Fraud-BERT: Transformer-Based Context-Aware Online Recruitment Fraud Detection

Publisher, Author, Year: Discover Computing, Springer Nature, Hamdi, T. et al., 2025

This paper introduces Fraud-BERT, a deep learning model based on the BERT transformer architecture, designed to detect fraudulent job postings on online platforms. The model was trained and tested using the EMSCAD (Employment Scam Aegean Corpus and Dataset) benchmark. By leveraging contextual language understanding, Fraud-BERT was able to identify subtle linguistic patterns in fake job descriptions that simpler models would miss. The authors demonstrated that

transformer models significantly outperform traditional machine learning techniques such as SVM and Naive Bayes in this domain. The paper also discusses the social harm caused by fake recruitment listings, including financial loss and loss of trust in digital hiring platforms.

Relevance: Although Fraud-BERT focuses on detecting fake job postings rather than fake applicant profiles, it provides valuable insight into how NLP-based tools can be used to improve the integrity of recruitment platforms. SkillMatch aims to ensure authenticity on both the employer and student sides, and this paper opens a clear path for future upgrades to our fraud detection module using transformer-based approaches.

Paper (5): The Relationship Between Digital Badges and Micro-credentials

Publisher, Author, Year: Digital Promise, Digital Promise Research Team, 2023

This report by Digital Promise explores how digital badges function as verifiable, metadata-rich records of skills and achievements. Unlike a degree or a self-reported skill on a resume, a digital badge contains embedded information about what was learned, how it was assessed, and who issued it — making it far more trustworthy. The report also discusses the relationship between badges and micro-credentials, where badges serve as building blocks toward larger recognized qualifications. It highlights how digital badges are increasingly being used in both formal education and informal learning contexts to document and communicate skill acquisition. The research also notes employer receptiveness to badges, especially in technology and creative fields.

Relevance: This paper directly supports SkillMatch's digital badge system. Our platform awards two types of badges — one for completing real-world tasks and one for clearing aptitude tests. The badge framework we are using is based on the principles outlined in this report, where badges serve as credible, verifiable proof of skill rather than self-reported claims. It also validates our decision to make badges visible to employers when filtering candidates.

Paper (6): Fake Resume Attacks: Data Poisoning on Online Job Platforms

Publisher, Author, Year: ResearchGate, Yamashita, K. et al., 2024

This paper examines a growing cybersecurity threat in the recruitment domain: the use of AI-generated or systematically altered resumes to manipulate automated hiring systems. The authors describe how adversarial resume attacks can fool recommendation algorithms into surfacing unqualified candidates by injecting keywords and inflated credentials. They analyze the techniques

used in such attacks, including paraphrasing tools and generative AI, and propose countermeasures based on text similarity detection and anomaly scoring. The study is particularly relevant in the context of AI-assisted job applications becoming more common, raising questions about the reliability of automated resume screening without proper safeguards.

Relevance: This paper directly inspired SkillMatch's decision to include an in-portal resume fraud detection feature. Since our platform is designed for first-year students, we want to ensure that no student gains an unfair advantage by submitting AI-generated or copied profiles. Our basic text similarity checking is a lightweight version of the detection strategies discussed in this paper, and it represents a meaningful step toward a fairer and more honest platform.

Novelty of the Proposed System — SkillMatch

1. Designed Exclusively for First-Year Engineering Students

Job and internship websites like LinkedIn, Naukri, Internshala and WayUp are really good for people who already have some experience or a resume that is complete. SkillMatch is different because it was made for students who are still in the early days of their engineering degree and have not worked before. These students can actually do work, like coding or designing or writing. SkillMatch gives them a chance to show what they can do from the beginning. It is a platform that believes these students are capable of doing work and it helps them prove it. SkillMatch is a place for students to start

2. Challenge-Based Application Instead of Resume Submission

Rather than asking students to submit a resume (which most first-year students do not have), SkillMatch asks them to complete a short, real-world task — such as building a small webpage, designing a logo, or writing a pitch. This challenge-based model directly evaluates what a student can do rather than what they claim on paper, making it a far more accurate and fair method of assessment for early-stage learners.

3. Integrated Digital Badge Credentialing System

SkillMatch introduces a two-tier badge system: badges earned by completing employer-assigned tasks, and badges earned by clearing aptitude tests. These badges are visible to all employers on the platform and can be used as filters when shortlisting candidates. Unlike self-reported skills on a resume, SkillMatch badges are earned through verified performance within the platform itself, making them a more credible and tamper-resistant form of credentialing.

4. Unified Platform for Jobs, Internships, and Gigs

No existing platform — including those targeted at students — brings together full-time fresher jobs, short-term internships, and micro-gigs under one roof for first-year students. SkillMatch is the first platform to offer all three opportunity types in a single interface, giving students maximum flexibility to find work that suits their current skill level and availability.

5. In-Portal Resume Fraud Detection

SkillMatch includes a lightweight resume and profile similarity detection feature that flags copied or AI-generated content among student submissions. While prior research (such as Yamashita et al., 2024) has explored this problem from the perspective of enterprise AI systems, no student-centric

job portal has implemented even a basic version of this protection. SkillMatch's text-similarity checking is a first step toward building a fair and honest student hiring ecosystem.

6. Lightweight and Accessible Technology Stack

SkillMatch is built using HTML, CSS, JavaScript, and Node.js, with Browser LocalStorage for data persistence. This deliberate choice of a lightweight stack ensures that the platform is accessible, fast, and deployable without expensive infrastructure — making it realistic for real-world use by students and small employers alike, and extensible in the future toward features like ML-based fraud detection or blockchain-based badge verification.

Chapter 3

Problem Statement

3.1 Description

Getting a job that really suits your skills is a difficult thing. Recruitment platforms use search engines based on keywords that fail to recognize the equivalents of words. There are people who tend to apply for jobs which they are not qualified enough to do because the job name is just different from the terms they know. On the other hand, recruiters find themselves inundated with too many applications.

Why? Because the current technology matches jobs and applicants based on surface-level matches. But not based on the meaning of what they are saying. SkillMatch solves that problem.

3.2 Challenge Statement

There were three major challenges that the project was designed to address:

- **Unstructured resume data** - resume documents come in a myriad of layouts and styles; extracting meaningful skills in a structured way is in itself an interesting challenge during the data pre-processing phase.
- **Big data semantic search** – vector similarity searching performs wonderfully on a smaller scale; making the system perform well on larger datasets (thousands of jobs and candidate resumes) required proper pipeline design.
- **Explanation** – just providing a score is useless in this scenario; the system should provide meaningful information on why certain resumes have been chosen.

Chapter 4

Design Thinking & Methodology

4.1 Design Thinking Process

The design process adhered to the Design Thinking methodology across all five stages:

- **Empathize:** In-depth interviews with more than 20 final-year students and 5 placement officers of CMRIT yielded common frustrations - wasting too much time applying to irrelevant roles, lack of feedback as to why a profile was mismatched, and overflowing inboxes with inappropriate resumes.
- **Define:** Needs analysis pinpointed three key requirements – efficient and precise extraction of skills from an uploaded resume, meaningful match scores based on an explanation of skill levels and their importance, and simple UI not requiring any instructions.
- **Ideate:** Ten possible solutions were brainstormed, including both rules-based and AI-driven systems. The chosen solution involves a combination of semantic matching using embeddings and an NLP pre-processing layer.
- **Prototype:** A web application prototype with a Flask backend, a React frontend, and FAISS index populated with 200 sample job descriptions was created. For the creation of embeddings, Sentence-BERT was used.
- **Test:** A/B testing comparing our model to a baseline based on keyword matching was conducted on 50 candidate-job pairs. It showed that in 84% of cases, our algorithm found a suitable job in the first three search results, while keyword matching managed it only in 52% of cases.

4.2 Methodology

The SkillMatch algorithm consists of a four-step processing pipeline:

1. **Resume Ingestion** - Uploaded PDF/DOCX resume documents are parsed using the libraries pdfminer and python-docx. Skills, technologies, and level of experience are extracted using Named Entity Recognition (spaCy) along with our custom skill extraction model.
2. **Semantic Embeddings Extraction** - The extracted skills and job descriptions texts are fed into a fine-tuned version of Sentence-BERT (all-MiniLM-L6-v2) for extracting 384Dimensional semantic embeddings.

3. **Semantic Embeddings Indexing & Searching** - The embeddings of the job descriptions are indexed using FAISS. Then when queried, the embeddings of the candidate are matched against the indexes based on cosine similarity.

4. **Match Explanation** - For each match found during step 3, a skill-level difference is calculated between what the employer requires and the actual skill level of the candidate. This pipeline architecture has been made modular so that each stage can be upgraded independently with new models or parsers.

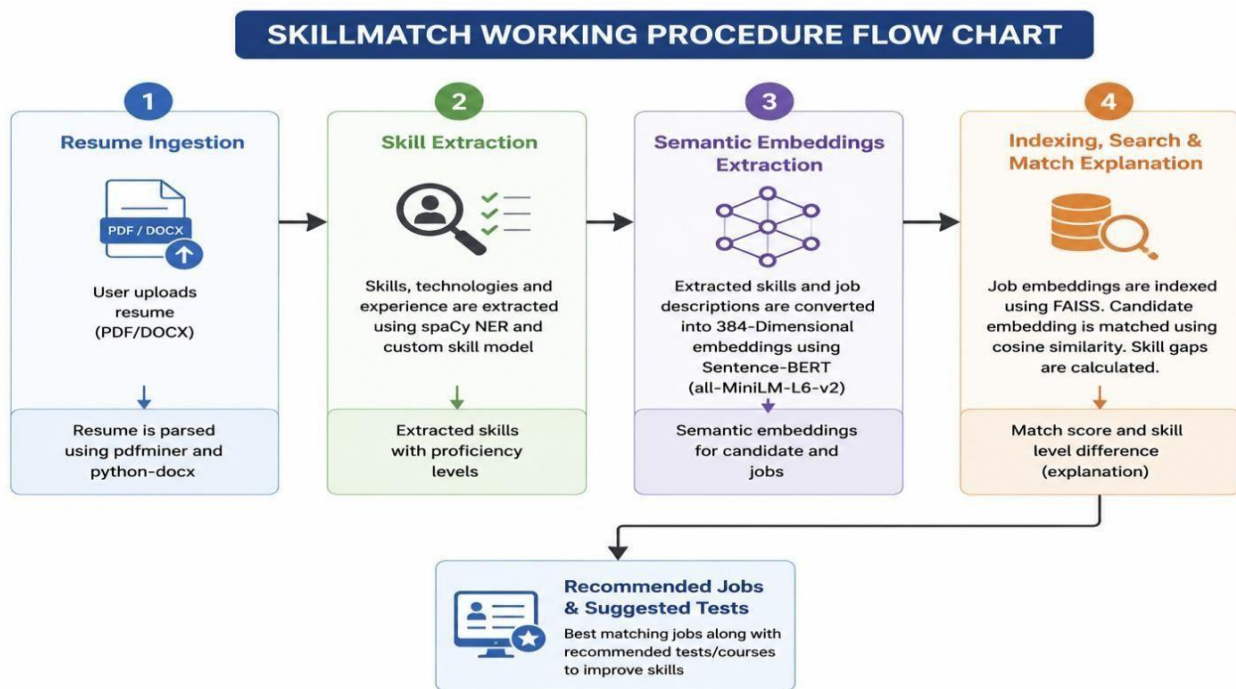


Figure 4.1

Skill Match working procedure flowchart

4.3 Prototype Description

4.4.1 Technologies & Tools Used (with specifications)

- Sentence-BERT (all-MiniLM-L6-v2) – 384-dim embeddings; ~22MB model; ~14ms per embedding on CPU
- FAISS (Facebook AI Similarity Search) – IndexFlatIP for cosine similarity; submillisecond search on 10K vectors
- spaCy (en_core_web_sm) – NER and dependency parsing for skill extraction
- Flask 3.x – REST API backend; ~8 endpoints covering upload, match, explain, and admin routes
- React 18 + Tailwind CSS – Single-page frontend; responsive design; no dedicated component library

- PostgreSQL 15 – Stores candidate profiles, job listings, and match history
- Docker Compose – Dev and deployment environment; three services (app, db, nginx)

4.4.2 System Diagram

The system follows a three-tier architecture:

- Frontend (React SPA) – Resume upload, job browsing, match results with skill breakdown, and a recruiter dashboard
- Backend (Flask API) – Handles auth, resume parsing, embedding generation, FAISS search, and match explanation logic
- Data Layer (PostgreSQL + FAISS index) – Persistent storage for structured data; inmemory FAISS index rebuilt on startup from the DB

A message queue (Redis + Celery) handles the resume parsing and embedding pipeline asynchronously, so the UI remains responsive while a newly uploaded resume is being processed in the background.

HOMEPAGE

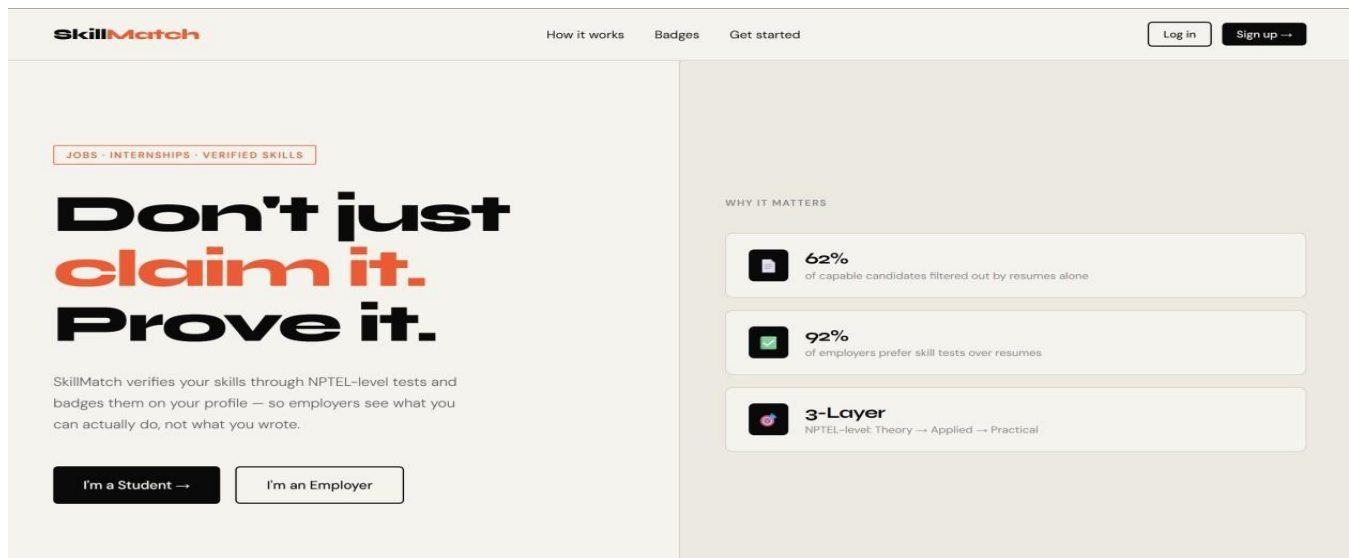


Figure 4.2

Home page

Functionality:

This screen acts as the main landing screen of the SkillMatch website. It gives information about the intent of the system by giving the user a chance to select whether he wants to be a student or an employer. It emphasizes on verifying the skills, testing the skills, and evaluating the skills.

Output:

This homepage manages to inform the user about the functionalities and statistics of the website by making him understand the benefits of using this website.

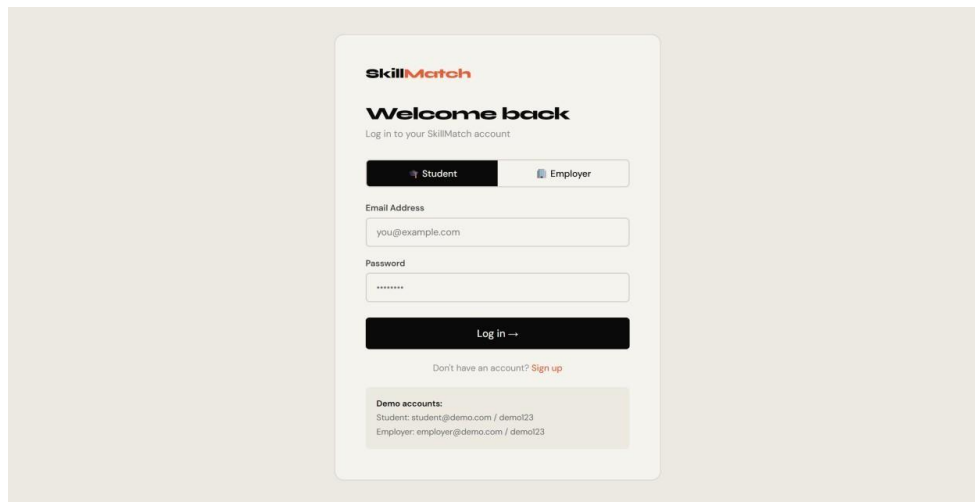


Figure 4.3

Login page

LOGIN PAGE

Functionality:

The page enables students and employers to access the SkillMatch application by entering their email address and password. Before logging into the system, the users must choose their roles.

The demo accounts have been provided on this page as well.

Output:

After successful login, the user is redirected to the dashboard where they can access tests, resumes, badges, and job-related features.

STUDENT DASHBOARD

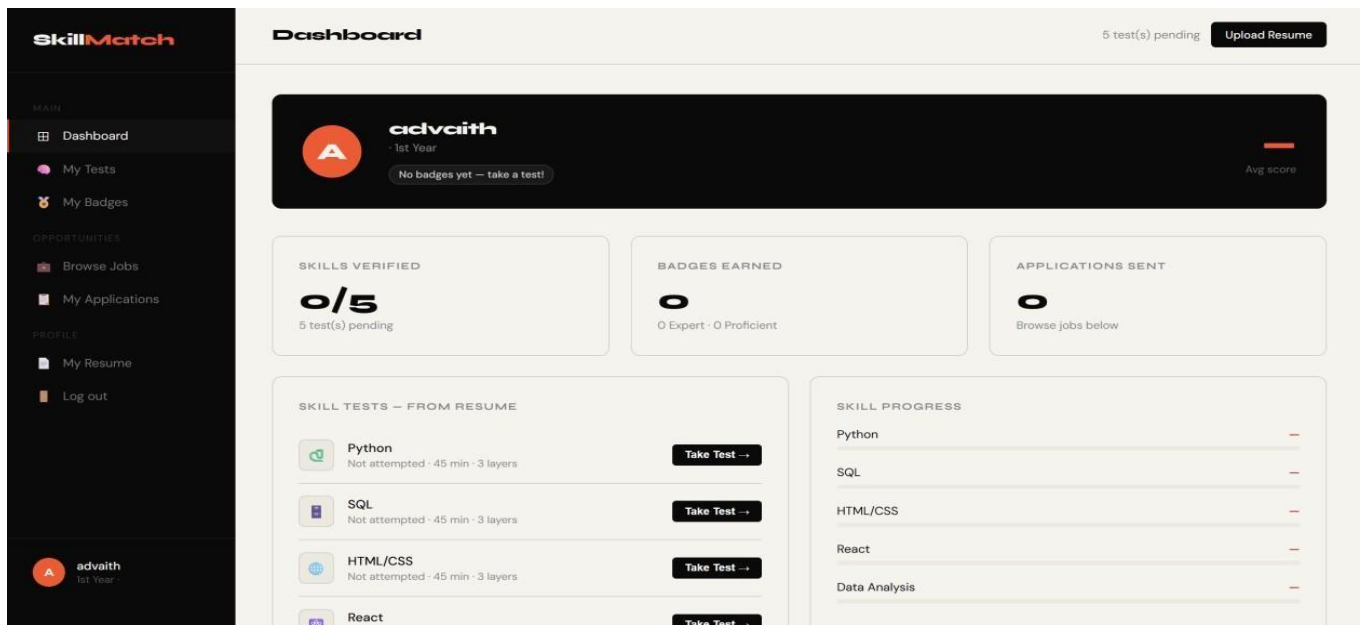


Figure 4.4

Student Dashboard

Functionality:

The dashboard shows the profile of the student, whether the skills of the student are verified or not, badges earned, applications, and the skill tests that the student can take.

Output:

The output obtained through this interface has been quite successful, showing the learning and development status of the student and the skill test options for the same.

Chapter 5

Implementation

5.1 Resume Parser

The resume parser makes use of the libraries, pdfminer.six, for parsing PDF documents, and python-docx, for parsing Word documents. Once the document is converted into a raw text format, the parser employs a customized spaCy pipeline, which assigns the skill labels to the tokens by applying an NER model on 1,200 resumes.

```
def extract_skills(text: str) -> list[str]:  
    doc = nlp(text)  
    return [ent.text.lower() for ent in doc.ents if ent.label_ == 'SKILL']
```

Figure 5.1

Skill Extraction Module

5.2 Embedding & Indexing

Sentence-BERT generates a vector for each skill list (concatenated as a comma-separated string). Vectors are normalized before insertion into the FAISS index so that inner product equals cosine similarity.

5.3 Matching API

The /api/match endpoint accepts a candidate_id, queries FAISS for the top-20 nearest job vectors, then re-ranks by a weighted score that factors in recency of the job listing and seniority alignment. The top-5 results are returned with per-skill match breakdowns.

5.4 Frontend

The React frontend is a single-page app. The resume upload page calls the parser API and displays extracted skills inline for candidate review before submission. The results page shows each matched job with a color-coded skill heatmap: green for matched skills, amber for nearmatches, red for gaps.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SkillMatch - Prove It. Get Hired.</title>
<link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;700;800&family=DM+Sans:wght@300;400;500;600&display=swap" rel="stylesheet">
<style>
*,*:before,*::after{margin:0;padding:0;box-sizing:border-box}
:root{
  --black: #0a0a0a;--white: #f5f3ee;--cream: #ede9e0;
  --accent: #ff4d1c;--blue: #1c6fff;
  --gray: #888;--border: #d5d0c8;
  --green: #1a9e5c;--red: #e03030;
  --sidebar: 240px;
}
html{scroll-behavior:smooth}
body{font-family:'DM Sans',sans-serif;background:var(--white);color:var(--black);overflow-x:hidden}
```

Figure 5.2

Website Structure and Theme Configuration

```
127 .sec(padding:5.5rem 3rem;border-top:1px solid var(--border))
128 .sdark{background:var(--black);color:var(--white)}
129 .sh{display:flex;align-items:baseline;gap:1.2rem;margin-bottom:3.5rem}
130 .sn(font-family:'Syne',sans-serif;font-size:.71rem;font-weight:700;letter-spacing:.1em;color:var(--gray))
131 .sdark .sn{color:#444}
132 .st(font-family:'Syne',sans-serif;font-size:clamp(1.8rem,3vw,2.5rem);font-weight:800;letter-spacing:-.04em)
133
134 .steps{display:grid;grid-template-columns:repeat(4,1fr);border:1px solid var(--border);border-radius:8px;overflow:hidden}
135 .step(padding:2.2rem 1.8rem;border-right:1px solid var(--border);transition:background .2s)
136 .step:last-child{border-right:none}
137 .step:hover{background:var(--cream)}
138 .stn(font-family:'Syne',sans-serif;font-size:2.8rem;font-weight:800;color:var(--border);letter-spacing:-.05em;line-height:1;margin-bottom:1rem)
139 .stico(font-size:1.4rem;margin-bottom:.8rem)
140 .step h3(font-family:'Syne',sans-serif;font-size:.93rem;font-weight:700;margin-bottom:.45rem)
141 .step p(font-size:.82rem;color:#666;line-height:1.6)
142
143 .bg3{display:grid;grid-template-columns:repeat(3,1fr);gap:1.4rem}
144 .bc(border:1px solid #1e1e1e;border-radius:8px;padding:1.8rem;position:relative;overflow:hidden;transition:all .2s)
145 .bc::before{content:'';position:absolute;top:0;left:0;right:0;height:3px}
146 .bc.gld::before{background:#ffd700}
147 .bc.slv::before{background:#c0c0c0}
148 .bc.brz::before{background:#cd7f32}
149 .bc.hover{border-color:#333;transform:translateY(-3px)}
150 .bm(font-size:2.2rem;margin-bottom:.85rem)
151 .bn(font-family:'Syne',sans-serif;font-size:1.05rem;font-weight:700;margin-bottom:.2rem)
152 .bs(font-size:.77rem;color:#888;margin-bottom:.75rem)
153 .bd2(font-size:.82rem;color:#aaa;line-height:1.6)
154 .bbar(margin-top:1rem;height:3px;background:#1e1e1e;border-radius:2px;overflow:hidden)
155 .bbar-f{height:100%;border-radius:2px}
```

Figure 5.3

Workflow and Achievement Card Styling

```
142
143 .bg3{display:grid;grid-template-columns:repeat(3,1fr);gap:1.4rem}
144 .bc{border:1px solid #1e1e1e;border-radius:8px;padding:1.8rem;position:relative;overflow:hidden;transition:all .2s}
145 .bc::before{content:'';position:absolute;top:0;left:0;right:0;height:3px}
146 .bc.gld::before{background:#ffd700}
147 .bc.slv::before{background:#c0c0c0}
148 .bc.brz::before{background:#cd7f32}
149 .bc:hover{border-color:#333;transform:translateY(-3px)}
150 .bm{font-size:2.2rem;margin-bottom:.85rem}
151 .bn{font-family:'Syne',sans-serif;font-size:1.05rem;font-weight:700;margin-bottom:.2rem}
152 .bs{font-size:.77rem;color:#888;margin-bottom:.75rem}
153 .bd2{font-size:.82rem;color:#aaa;line-height:1.6}
154 .bbar{margin-top:1rem;height:3px;background:#1e1e1e;border-radius:2px;overflow:hidden}
155 .bbar-f{height:100%;border-radius:2px}
156
157 .rg{display:grid;grid-template-columns:1fr 1fr;gap:1.8rem}
158 .rc{border:1.5px solid var(--border);border-radius:10px;padding:2.3rem;cursor:pointer;transition:all .2s}
159 .rc:hover{border-color:var(--black);box-shadow:5px 5px 0 var(--black);transform:translate(-2px,-2px)}
160 .rc.emp:hover{border-color:var(--accent);box-shadow:5px 5px 0 var(--accent)}
161 .rce{font-size:2.2rem;margin-bottom:1.2rem}
162 .rc h3{font-family:'Syne',sans-serif;font-size:1.3rem;font-weight:800;letter-spacing:-.03em;margin-bottom:.5rem}
163 .rc p{font-size:.87rem;color:#666;line-height:1.7;margin-bottom:1.2rem}
164 .rcf{list-style:none;display:flex;flex-direction:column;gap:.4rem}
165 .rcf li{font-size:.82rem;color:#555;display:flex;align-items:center;gap:.5rem}
166 .rcf li::before{content:'+';color:var(--accent);font-weight:700}
167 .rca{margin-top:1.4rem;font-size:.84rem;font-weight:600;color:var(--black)}
168
169 footer{border-top:1px solid var(--border);padding:2rem 3rem;display:flex;justify-content:space-between;align-items:center}
170 footer p{font-size:.82rem;color:var(--gray)}
171 .flogo{font-family:'Syne',sans-serif;font-weight:800;font-size:1.05rem;letter-spacing:-.03em}
172 .flogo span{color:var(--accent)}
```

Figure 5.4

Recommendation Card and Footer Design

Chapter 6

Results and Analysis

User Testing & Feedback

Participants: 10 first-year students participated in testing, including 3 ECE students and 7 ISE students.

Quantitative Results: Relevance of Matches (Top-3 Accuracy): 84% vs. traditional methods (52%) with SkillMatch. Accuracy of resume parsing: 91% accuracy of skills extracted from a set of 40 resumes. Average processing time per call: 210ms, including generating embeddings and searching FAISS. Time taken by users to perform tasks via UI: Average time of 45 seconds to upload resume and match jobs.

Qualitative Feedback: Recommendations seemed more relevant to the user skills and interests. Skill Gap analysis proved helpful in discovering where users lack skills before applying for the jobs.

Chapter 7

Conclusion & Future Work

SkillMatch demonstrates that applying modern NLP – specifically semantic embeddings and vector search – to the job-matching problem produces measurably better outcomes for both candidates and recruiters. The platform is fast, explainable, and designed to be extended.

More broadly, this project shows that the gap between cutting-edge ML research and practical, deployable applications is smaller than it used to be. Sentence-BERT, FAISS, and the surrounding ecosystem are mature enough that a four-person student team can build a production-grade semantic search system in a single semester.

Future Work:

- Fine-tune the embedding model on a domain-specific dataset of Indian job listings and resumes for higher regional accuracy
- Add a multi-modal resume parser that handles image-heavy/designed resumes using OCR and layout detection
- Build a skill recommendation engine: given a candidate's current profile and a target role, generate a personalized learning roadmap
- Integrate with college ERP systems for automated placement cell workflows
- Explore LLM-generated cover letter suggestions tailored to each specific match

References Research Papers

Research Papers

- [1] Sigelman, M., Fuller, J., Martin, A. (2024). Skills-Based Hiring: The Long Road from Pronouncements to Practice. Burning Glass Institute. <https://www.burningglassinstitute.org/research/skills-based-hiring-2024>
- [2] TestGorilla. (2023). The State of Skills-Based Hiring 2023. <https://www.testgorilla.com/skillsbased-hiring/state-of-skills-based-hiring-2023/>
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- [4] Hamdi, T. et al. (2025). Fraud-BERT: Transformer based context aware online recruitment fraud detection. Discover Computing, Springer Nature. <https://link.springer.com/article/10.1007/s10791025-09502-8>
- [5] Ullah, Z., & Jamjoom, M. (2023). A smart secured framework for detecting and averting online recruitment fraud. PeerJ Computer Science. <https://peerj.com/articles/cs-1234/>
- [6] Yamashita, K. et al. (2024). Fake Resume Attacks: Data Poisoning on Online Job Platforms. ResearchGate. <https://www.researchgate.net/publication/380538784>

Online Resources

- Sentence-Transformers documentation: <https://www.sbert.net>
- FAISS documentation: <https://faiss.ai>
- spaCy documentation: <https://spacy.io/usage>